Session 3: Git and $\LaTeX$ in VS Code

Version Control and the Integrated Research Environment

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Where We Are

Session 1: What Claude Code can do. How it works. CLAUDE.md and plan mode.

Session 2: The core workflow. Data cleaning, estimation, robustness, LaTeX tables.

Session 3: Building infrastructure around the workflow.

Today you will:

1. Learn the minimum viable git workflow for agentic coding
2. Bring your paper into the same project directory as your code
3. Use Claude Code to write, compile, and version-control a LaTeX document

Session 2 taught you to verify Claude Code's work. Today you get the tools to manage it.

Check-In: Claude Code in Your Work

Since Sessions 1 and 2, have you tried Claude Code on your actual research?

If yes: What did you use it for? Data cleaning? Writing code? Something else?

If no: What's holding you back? Unclear how? Don't trust it yet? Wasn't the right tool?

Check-In: Claude Code in Your Work

- **Current usage:** Are you using Claude Code in your work at all?
- **What for:** What kind of tasks?
- **Pain points:** What's been frustrating or hard?
- **Prospective:** What tasks have you been thinking about trying it on but haven't yet?
- **Confusion:** What's still unclear about how it works or when to use it?
- **Trust:** Are you confident in the code it produces?

Git is research insurance,
not software engineering ceremony.

You need exactly four commands.

What Are Git and GitHub?

Git is a tool that tracks every change to every file in a project.
It runs on your computer. It is free and open source.

Think of it as "track changes" for your entire project directory — not just one document, but every file: code, data descriptions, LaTeX files, everything.

GitHub is a website that hosts git projects online.

- It is where open-source code, replication packages, and increasingly papers live
- Git works entirely on your machine. GitHub is optional — it adds backup, sharing, and collaboration
- Today we only use git locally. GitHub comes in Session 4.

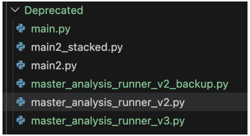
The Key Idea

Git lets you save snapshots of your project at any point. You can always see what changed between snapshots, and go back to any

Why Researchers Should Care About Git

Without git, you have probably done this:

- `analysis_v2.py`, `analysis_v2_final.py`, `analysis_v2_final_REAL.py`
- "Which version of the cleaning script produced these results?"
- "I changed something last week and now the code is broken"



With git:

- One file, full history. Every version is recoverable.
- You can see exactly what changed, when, and why.
- You can undo any change — even from weeks ago.

Git is useful for any research project. But when you are working with an agent that modifies your files directly, it goes from useful to essential.

Why Git Matters for Agentic Coding

The problem: Claude Code modifies your files directly.

- It might change a file you did not ask it to change
- It might overwrite working code with something broken
- It might make 10 changes when you only wanted 3

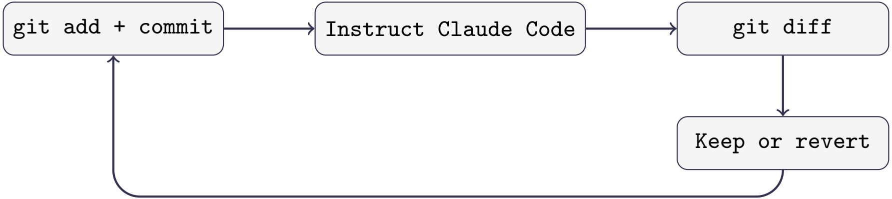
Without git: You try to manually undo changes. You are not sure what the file looked like before. You lose work.

With git: You see exactly what changed, line by line. You revert anything you do not want. You never lose work.

The Rule

Never let Claude Code touch your project without a recent commit to fall back on.

The Minimum Viable Git Workflow



Command	What it does
`git add -A && git commit -m "msg"`	Snapshot your current state
`git diff`	See what changed since last commit
`git checkout -- file.py`	Undo changes to a specific file
`git log --oneline`	See your commit history

Command 1: `git init`

What it does: Turns a regular folder into a git project.

```
git init
```

- You run this **once**, when you start a new project
- It creates a hidden `.git/` folder that stores your project's history
- Nothing else changes — your files stay exactly as they are

Think of it as saying: "I want to start tracking this folder."

Note

If you cloned this course repository, git is already initialized. You can check by running `git status` — if it prints something, you are already set up.

Command 2: `git add`

What it does: Selects which files to include in your next save point.

```
git add -A
```

- `-A` means "add everything that changed" — new files, modified files, deleted files
- This does not save anything yet. It just marks what will be included in the next commit.

You can also add specific files instead of everything:

```
git add scripts/analysis.py
```

Think of it as putting files in a box before sealing it. `git add` fills the box; `git commit` seals it.

Command 3: `git commit`

What it does: Saves a snapshot of everything you added.

```
git commit -m "Clean merge script, verified row counts"
```

- `-m` stands for "message" — a short note describing what this snapshot contains
- The message goes in quotes. Write something your future self will understand.
- After committing, you have a permanent save point you can always return to.

In practice, you can always run add and commit together (but you don't have to):

```
git add -A && git commit -m "your message"
```

The `&&` just means "run the second command after the first one finishes."

Command 4: `git diff`

What it does: Shows you exactly what changed since your last commit.

```
git diff
```

The output looks like the diff view you already know from VS Code:

- Lines starting with + were added (green in VS Code)
- Lines starting with – were removed (red in VS Code)
- Everything else is unchanged context

When to use it: After Claude Code makes changes, run `git diff` to see *everything* it touched — not just the file it told you about.

Tip

You can also ask Claude Code: "Run git diff and explain each change you made."

Command 5: `git log`

What it does: Shows you the history of all your commits.

```
git log --oneline
```

Output looks like:

```
a3f7b21 Add robustness checks
e5c9d04 Clean merge script, verified row counts
b1a2c33 Initial commit
```

- Each line is one commit — one save point
- The letters and numbers on the left are the commit ID
- The message on the right is what you wrote with `-m`
- Most recent commits are at the top

This is your project's timeline. You can always see what you did and when.

Command 6: `git checkout`

What it does: Undoes changes by restoring files to a previous committed state.

Undo uncommitted changes to one file:

```
git checkout -- scripts/analysis.py
```

Undo all uncommitted changes:

```
git checkout -- .
```

Go back several commits (rewind the whole project):

```
git log --oneline # find the commit you want
git checkout <commit-hash> # jump to that commit
git checkout main # come back to the latest
```

Or use `HEAD~N` to go back `N` commits from your current position:

```
git checkout HEAD~3 # go back 3 commits
```

`git checkout`: Your Undo Button

The mental model:

- `git checkout -- .` → undo *uncommitted* work
- `git checkout <hash>` → travel back to any previous commit
- `git checkout main` → return to the latest state

Recovering if something goes wrong:

```
git reflog # shows every action you've taken
```

Even changes you thought were gone are usually recoverable.

This all only works if you committed first

Checkout lets you jump between snapshots — but you need snapshots to jump to. That is why you always commit before instructing Claude Code.

Putting It Together: The Cycle

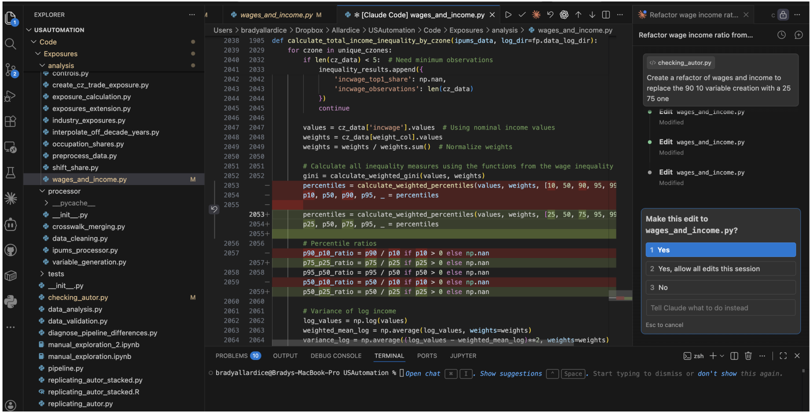
Before every interaction with Claude Code:

1. `git add -A && git commit -m "message"` — save your current state
2. Give Claude Code an instruction
3. `git diff` — see what it changed
4. Decide:
   - Happy? → `git add -A && git commit -m "message"`
   - Not happy? → `git checkout -- .`
5. Repeat

The Mindset Shift

Commits are not milestones. They are save points — like pressing Ctrl+S. Commit early, commit often, commit before every interaction with Claude Code.

You Have Already Seen This



Red = removed, green = added. You have been reading diffs since Session 1.

In VS Code: When working directly in the editor, you see the `git diff` visually before you accept the change. This shows exactly what Claude Code changed.

Git and Data: What Not to Track

Git is designed for **code and text files** — things that change line by line.
It is *not* designed for large data files.

Commit:

- Scripts (`.py`, `.R`, `.do`)
- LaTeX files (`.tex`, `.bib`)
- Documentation (`CLAUDE.md`)
- Small data files (under a few MB)

Do not commit:

- Large datasets (50MB+)
- Output you can regenerate
- Sensitive data (PII, API keys)

Rule of Thumb

Commit what a collaborator needs to *understand* your project. Skip what they could *regenerate* by running the code.

`.gitignore`: Telling Git What to Skip
`.claudeignore`: Telling Claude What to Skip

A `.gitignore`/`.claudeignore` file lists patterns for files git/claude should never track. Place it in the folder where you ran `git init`.

Paths are relative to that folder. Example:

```
my_project/ <- run "git init" here
  .gitignore <- goes here
  .claudeignore <- goes here
  code/
    analysis.py
  data/ <- put in .gitignore but NOT .claudeignore
    raw/
    processed/
  output/ <- put in .gitignore but NOT .claudeignore
```

Always include in .gitignore and .claudeignore:

```
.DS_Store
.env <- usually contains sensitive information
*.pyc
__pycache__/
*.aux
*.log
```

Then add: `data/raw/`, `*.csv.gz`, any large files.

.gitignore vs .claudeignore

.gitignore

```
# Folders
data/raw/
data/*/administrative/
output/

# Specific files
.env
.DS_Store

# Large Data Files
data_merged.csv
copy_of_internet.csv
output/

# Python cache (auto-generated, don't need)
__pycache__/
```

.claudeignore

```
# Sensitive files
.env
.DS_Store
NUCLEAR_CODES_DO_NOT_SHARE.txt
list_of_my_biggest_fears.docx

# Large files you dont want claude to waste time reading (bloats
      history, wastes context)
*.csv.gz

# Cache (auto-generated, eats token budget)
__pycache__/
```

Wildcards: * matches any characters.

- `*.pyc` = all `.pyc` files (anywhere in project)
- `data.*.csv` = `data.raw.csv`, `data.clean.csv`, etc.
- `analysis_*.py` = `analysis_v1.py`, `analysis_v2.py`, etc.

Folders must end with / (e.g., `output/`, `__pycache__/`)

So Where Does the Data Go?

Why git is bad for data:

- Git stores *every version* of every file. A 200MB CSV edited 10 times = 2GB of history.
- Git compares files line by line. Change one cell in a CSV and git sees the entire row as new.
- Large files slow down every git operation — commits, diffs, cloning.

For most research projects (data under a few GB):

- Keep data in a synced folder: **Dropbox**, **OneDrive**, **Google Drive**
- Your code lives in git. Your data lives next to it. Your `.gitignore` keeps them separate.
- Simple, familiar, and good enough for most social science workflows

For larger or more complex needs:

- **Object storage:** AWS S3, Google Cloud Storage — cheap, scalable
- **Data versioning:** DVC (Data Version Control) — git-like tracking for large files

You do not need these today. But if your data outgrows Dropbox, they exist.

Exercise Part 1: Choose Your Project

Option A: Your own research project

- Open your project folder in VS Code: File → New Window → File → Open Folder
- Use whatever code and data you have in that project

Option B: Practice sandbox

- On your Desktop, create a new folder: `mkdir ~/Desktop/session_2_git_practice`
- Copy Session 2 materials into it: `cp -r module_2/* ~/Desktop/session_2_git_practice/`
- Open it in VS Code: File → New Window → File → Open Folder → select `session_2_git_practice` on Desktop

Pick one and open it in VS Code.

Exercise Part 1: Initialize and Commit (10 min)

Now, in your chosen project directory:

1. **Create a** `.claudeignore` FIRST: In VS Code (File → New File → name it .claudeignore) or terminal (`echo ".env" > .claudeignore`). Add `.env` and any other sensitive files.
   Important (Option B): You will find a `.env` file with my real credentials. Exclude it immediately.
2. **Create a** `.gitignore` now: Ask Claude Code to help. Include large data files, regenerated output, and system files.
   Note: `.claudeignore` uses the same syntax.
3. **Initialize git:** `git init` then `git add -A && git commit -m "Initial commit"`
4. **Run:** `git log --oneline`

Exercise Part 2: The Cycle (25 min)

Do this **three times** (you decide what changes):

1. **Commit** your current state
2. **Instruct Claude Code** to make a change. Pick anything:
   - Add a control variable, refactor into functions, change sample restriction
   - Fix a bug, improve documentation, rename variables
   - Add a new analysis, create a visualization, write a summary
3. **Run** `git diff` and read what changed
4. **Decide:** keep (commit) or revert (`git checkout -- .`)

Goal: Try to revert at least once. See that you can always undo.

If you revert and regret it, you can recover the changes from your last commit using `git reflog` and `git checkout`.

Commit → Instruct → Diff → Decide

Git Debrief

Discussion:

- Did Claude Code change any files you did not expect?
- Was the diff easy to read? What was confusing?
- Did anyone revert a change? What went wrong?

The Habit

Commit → Instruct → Diff → Decide.

This is not optional when working with an agent that modifies your files. It is the minimum responsible workflow.

In Session 4, we will automate the commit step with a hook — so you never forget.

Part 3: The Integrated Research Environment

What if your paper lived in the same place
as your code and data?

This is the conceptual pivot of the course.

The Problem with Overleaf

Most researchers keep their paper in a separate tool.

- Code lives in a project directory (or on your Desktop)
- Data lives next to the code (maybe)
- The paper lives in Overleaf (or Word, or Google Docs)

What this means for Claude Code:

- It cannot read your paper
- It cannot check the methods section against your code
- It cannot update a table when the analysis changes
- It cannot commit changes to your paper alongside code changes

The Limitation

If Claude Code cannot see it, Claude Code cannot operate on it.
Your paper is the biggest piece of your project that is currently invisible.

The Solution: LaTeX in VS Code

Bring your paper into the project directory.

- Your `.tex` file lives alongside `scripts/` and `data/`
- Claude Code can read it, write it, and edit it — just like any other file
- Changes to the paper are tracked by git, just like code changes
- Coauthors review changes through version control, not "track changes"

What becomes possible:

- Claude Code writes a methods section and you verify it against the code
- The analysis changes → the table updates → the paper updates
- Every version of the paper is recoverable through git

The Point

This is not about LaTeX vs. Word. It is about bringing your paper into the agentic workflow.

LaTeX in VS Code: The Setup

What you need (should be installed from pre-work):

1. A TeX distribution: TeX Live (Linux/Windows) or MacTeX (Mac)
2. The **LaTeX Workshop** extension in VS Code

What you get:

- Syntax highlighting for `.tex` files
- One-click compilation to PDF
- PDF preview in a split pane — side by side with your code
- Error messages in the VS Code terminal

The workflow:
Edit `.tex` → Save → Auto-compile → PDF updates in the preview pane.

You never leave VS Code. Your paper is right next to your code.

LaTeX Workshop: Essential Commands

Three ways to compile:

Keyboard Shortcuts (Mac):

```
Cmd+Alt+B Compile to PDF
Cmd+Alt+V Open PDF preview
```

Keyboard Shortcuts (Windows/Linux):

```
Ctrl+Alt+B Compile to PDF
Ctrl+Alt+V Open PDF preview
```

Or click the green button:

Once the preview is open:

- Right-click the PDF preview tab and select `Move into Side Panel`
- Now you have code on the left, PDF on the right

Optional: Auto-compile on save

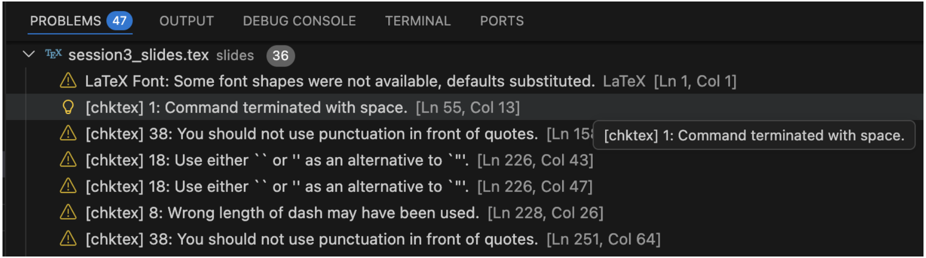
Add to VS Code settings: `"latex-workshop.latex.autoBuild.run": "onFileChange"`

LaTeX Errors and the Problems Pane

When you compile, LaTeX Workshop checks for errors and syntax issues. They appear in the **Problems** pane at the bottom of VS Code.

To view the Problems pane:

- Keyboard: `Ctrl+Shift+M` (or `Cmd+Shift+M` on Mac)
- Menu: `View` → `Problems`



Errors and warnings appear here with line numbers. Click any error to jump to that line in your `.tex` file.

Why All This Matters for Claude Code

The payoff: Everything is in one place. Your code, your data, your LaTeX document, and your compilation errors are all visible to Claude Code.

Claude Code can:

- **See the error:** Claude Code reads the Problems pane and helps you fix it
- **Write the LaTeX:** Describe what you want (a table, a figure reference, a citation) and Claude Code writes it
- **Check consistency:** Ask Claude Code to verify that your methods section matches your actual code
- **Iterate quickly:** Write → Compile → Show error to Claude Code → Revise → Repeat

The Workflow

Your `.tex` file is no longer separate from your research. It is part of your project, subject to the same verification and version control as your code.

LaTeX Basics (For Those New to It)

```
\documentclass{article}
\usepackage{booktabs}
\title{My Research Paper}
\author{Your Name}
\begin{document}
\maketitle
\section{Data Description}
We use a survey of 2,044 Polish adults...
\section{Results}
\input{output/robustness_table.tex}
\end{document}
```

Key idea: `\input{output/robustness_table.tex}` pulls in the table from Session 2. When the table updates, the paper updates.

Live Demo: Building a Paper from Scratch

What I'm going to show you:

I have a folder with pre-generated outputs from a fake analysis of *this course*:

- Summary statistics table (`.tex`)
- Regression results table (`.tex`)
- Four figures: enrollment, satisfaction, attrition, earnings effect

Starting from nothing, I'll use Claude Code to:

1. Create a blank `course_paper.tex` file
2. Write methods and data sections describing the study
3. Pull in the pre-generated tables with `\input{...}`
4. Embed the figures with `\includegraphics{...}`
5. Compile to PDF and fix any errors that come up

Watch how Claude Code reads the Problems pane and iterates without me copy-pasting errors.

Demo: The Full Loop

Claude Code writes a methods section → You verify it against the code → Compile to PDF → Commit with git

Everything happens in VS Code.

Everything is version-controlled.

Everything is visible to Claude Code.

Exercise Part 3: Build a LaTeX Document (30 min)

Choose your adventure:

- **Option A:** Use **your own work** — a current or recent research project with tables and figures you already have
- **Option B:** Use the **provided project** (`module_3/`) with FX data, summary stats, main table, and robustness table

Your goal: Build out two sections of a paper — **Data** and **Results**.

1. **Commit** your current state (the habit starts now)
2. **Data section:** Ask Claude Code to
   - Write text describing the data
   - Include one descriptive figure
     (Option B: the FX exchange rate over time, `fx_rate_figure.png`)

Exercise Part 3: Build a LaTeX Document (cont.)

3. **Results section:** Ask Claude Code to include three tables, each with a paragraph of text describing what it shows:
   - Summary statistics (`summary_stats.tex`)
   - Main results (`main_table.tex`)
   - Robustness (`robustness_table.tex`)
4. **Compile** to PDF, fix errors with Claude Code, then **diff and commit**
5. **Ask Claude Code** to check your text against the actual code. Does what you wrote match what the code does?

You are now using git and LaTeX together. Every change to the paper is tracked.

What You Built Today

1. **Git:** You have a safety net. Commit → instruct → diff → decide. You practiced reverting changes and saw that you can always go back.
2. **LaTeX in VS Code:** Your paper is now part of your project. Claude Code can see it, edit it, and check it against your code.
3. **The two together:** Every change to your paper is version-controlled, just like every change to your code. You used the git cycle on your `.tex` file.

The Bigger Picture

Your project directory now contains: data, code, output, and a paper — all under version control, all visible to Claude Code. This is the foundation for everything in Session 4.

Debrief

Discussion questions:

1. Did Claude Code change any files you did not expect? How did you handle it?
2. Did anyone revert a change? What went wrong?
3. When Claude Code wrote parts of the LaTeX document, what did you have to correct?
4. How does this compare to your current workflow — Overleaf, Word, separate directories?

Key Takeaway

The skill is not just using Claude Code. It is building an environment where Claude Code can do its best work — and where you can verify and undo everything it does.

Next Session

Session 4: Power Features — Hooks, MCPs, and Skills

- **Hooks:** Automate the commit step — every change Claude Code makes is tracked automatically
- **MCPs:** Connect Zotero — search your library and pull citations from inside Claude Code
- **Skills:** Reusable workflows — spec validation, robustness checks, and more

Before next session:

- Practice the commit-instruct-diff-decide cycle on your own project
- Try asking Claude Code to review a piece of your own code
- Make sure your Zotero account is set up (instructions in the setup guide)